

These ciphers replace letters in the plaintext with other letters or symbols, but the order in which the symbols occur remains the same. The **Caesar cipher** table below has a shift of 8 . this key is **CCS8** (*Caesar cipher shift 8*).

Plaintext	<i>A</i>	<i>B</i>	<i>C</i>	<i>D</i>	<i>E</i>	<i>F</i>	<i>G</i>	<i>H</i>	<i>I</i>	<i>J</i>	<i>K</i>	<i>L</i>	<i>M</i>	<i>N</i>	<i>O</i>	<i>P</i>	<i>Q</i>	<i>R</i>	<i>S</i>	<i>T</i>	<i>U</i>	<i>V</i>	<i>W</i>	<i>X</i>	<i>Y</i>	<i>Z</i>
Ciphertext	<i>S</i>	<i>T</i>	<i>U</i>	<i>V</i>	<i>W</i>	<i>X</i>	<i>Y</i>	<i>Z</i>	<i>A</i>	<i>B</i>	<i>C</i>	<i>D</i>	<i>E</i>	<i>F</i>	<i>G</i>	<i>H</i>	<i>I</i>	<i>J</i>	<i>K</i>	<i>L</i>	<i>M</i>	<i>N</i>	<i>O</i>	<i>P</i>	<i>Q</i>	<i>R</i>

2.Tic -Tac- Toe Cipher(TTTC.H)

A	B	C	J	K	L
D	E	F	M	N	O
G	H	I	P	Q	R

~~| | | |
|---|---|---|
| | S | |
| T | | U |
| | V | |~~

~~$$\begin{array}{ccccc} & & W & & \\ X & & \vdots & & Y \\ & & Z & & \end{array}$$~~

The phrase: "*The parrot knows*" would be encrypted as:

$$\square \in \langle \sqcup, \sqcap, \sqcup, \sqcap, \sqcup, \sqcap \rangle \quad \square \in \langle \sqcup, \sqcap, \sqcup, \sqcap \rangle$$

3. Affine Cipher(AC.M7.A11)

In an affine cipher each letter of the alphabet is converted to its numerical equivalent by a mathematical function using mod 26

STEP 1: begin by listing the alphabet and assigning a number, {0, 1, 2, 3, ... 23, 24, 25} to each.

a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z
0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25

STEP 2: select a multiplier. For this cipher let's multiply by 7 (the multiplier must not contain a factor of 26) and then add 11. Multiply each number in the table by 7 then add 11 and place the result in the next row (third row) of your table.

alphabet	a	b	c	d	e	f	g	h	i	j	k	l	m
convert to numbers	0	1	2	3	4	5	6	7	8	9	10	11	12
X by 7 then +11	11	18	25	32	39	46	53	60	67	74	81	88	95

alphabet	n	o	p	q	r	s	t	u	v	w	x	y	z
convert to numbers	13	14	15	16	17	18	19	20	21	22	23	24	25
X by 7 then +11	102	109	116	123	130	137	144	151	158	165	172	179	186

STEP 3: reduce each number in the third row to mod 26 and place in the fourth row of the table.

alphabet	a	b	c	d	e	f	g	h	i	j	k	l	m
convert to numbers	0	1	2	3	4	5	6	7	8	9	10	11	12
X by 7 then +11	11	18	25	32	39	46	53	60	67	74	81	88	95
convert to mod 26	11	18	25	6	13	20	1	8	15	22	3	10	17

alphabet	n	o	p	q	r	s	t	u	v	w	x	y	z
convert to numbers	13	14	15	16	17	18	19	20	21	22	23	24	25
X by 7 then +11	102	109	116	123	130	137	144	151	158	165	172	179	186
convert to mod 26	24	5	12	19	0	7	14	21	2	9	16	23	4

STEP 4: convert each number in the fourth row to the corresponding letter of the alphabet and place in the fifth row of the table.

alphabet	a	b	c	d	e	f	g	h	i	j	k	l	m
convert to numbers	0	1	2	3	4	5	6	7	8	9	10	11	12
X by 7 then +11	11	18	25	32	39	46	53	60	67	74	81	88	95
convert to mod 26	11	18	25	6	13	20	1	8	15	22	3	10	17
convert to alphabet	L	S	Z	G	N	U	B	I	P	W	D	K	R

alphabet	n	o	p	q	r	s	t	u	v	w	x	y	z
convert to numbers	13	14	15	16	17	18	19	20	21	22	23	24	25
X by 7 then +11	102	109	116	123	130	137	144	151	158	165	172	179	186
convert to mod 26	24	5	12	19	0	7	14	21	2	9	16	23	4
convert to alphabet	Y	F	M	T	A	H	O	V	C	J	Q	X	E

Now we have our key **AC.M7.A11** (*affine cipher multiply by 7 then add 11*)

a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z
L	S	Z	G	N	U	B	I	P	W	D	K	R	Y	F	M	T	A	H	O	V	C	J	Q	X	E

Thus "*Pirate treasure*" is encrypted as "MPALON OANLHVAN".

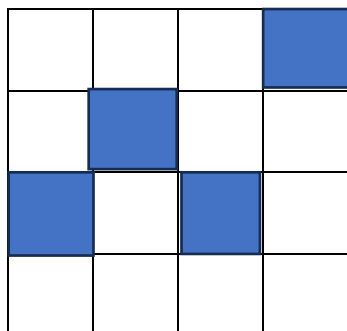
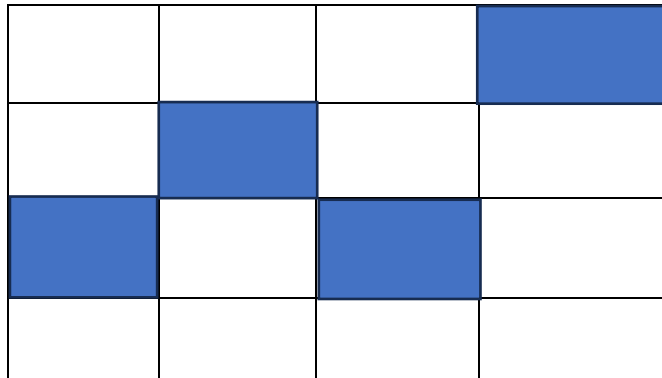
3. Grille Cipher (GC.S)

One transposition cipher, the **grille cipher**, positions the ciphertext in a square grid. The cipher key is a square grid with cut-out squares called the grille. To decipher the message you place the grille over the ciphertext square grid and read off the letters showing through the cut-out squares as they appear, left to right top to bottom.

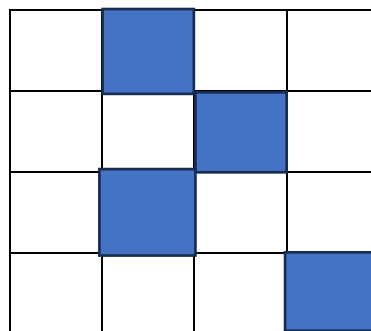
Next, you rotate the grille 90° clockwise and again read the letters.

Rotate the grille 180° and repeat.

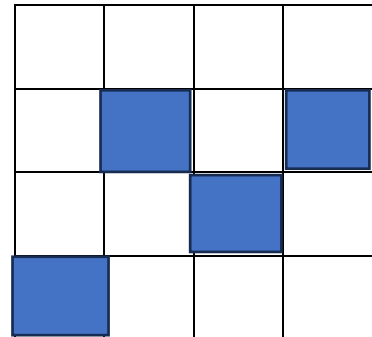
Finally rotate the grille 270° and repeat.



0° rotation



90° rotation



180° rotation